Your grade: 96%

Your latest: **96%** • Your highest: **96%** • To pass you need at least 80%. We keep your latest score.

Next item

Instructions

1. What is the primary advantage of applying hidden Markov models (HMMs) in the context of detecting metamorphic malware, which frequently alters its code structure? **1 / 1 point**

- HMMs require no training data to be effective.
- HMMs can model the statistical properties of sequences, making it easier to identify variations in metamorphic malware.
- HMMs can detect malware in real-time with zero false positives.
- HMMs are designed to target and eradicate the malware's payload.

Nice work

Correct! HMMs excel at modeling the statistical properties of sequences, which helps in detecting variations in metamorphic malware despite code changes.

2. Which methods are commonly employed to detect metamorphic malware, which continuously alters its code to evade detection? (Select all that apply) **0.8 / 1 point**

- [x] Control flow graph analysis

Nice work

Correct! Control flow graph analysis examines the execution paths within malware, which helps in identifying metamorphic code changes.

- [ ] Pattern matching
- [x] Static code analysis

This should not be selected

Incorrect. Static code analysis often struggles with metamorphic malware due to its dynamic code transformations.

- [x] Machine learning algorithms

Nice work

Correct! Machine learning algorithms can learn from patterns and anomalies, making them useful for detecting metamorphic malware.

- [x] Behavioral analysis

Nice work

Correct! Behavioral analysis monitors runtime behavior to detect anomalies, making it effective against metamorphic malware.

3. In malware detection, which technique is particularly suited for identifying polymorphic malware, which changes its appearance to avoid detection? **1 / 1 point**

- Heuristic analysis
- Signature-based detection
- Port scanning
- Sandboxing

Nice work

Correct! Heuristic analysis evaluates the behavior and patterns of malware, making it effective for detecting polymorphic variations.

4. Imagine a cybersecurity team is engaged in the ongoing battle against evolving malware threats. What are some notable aspects of this arms race between attackers and defenders? (Select all that apply) **1 / 1 point**

- [x] Attackers constantly develop new evasion techniques.

Nice work

Correct! Attackers continuously innovate new methods to evade detection and bypass security measures.

- [x] Defenders use a combination of heuristic and behavioral analysis.

Nice work

Correct! Defenders employ a diverse range of techniques, including heuristic and behavioral analysis, to counteract evolving malware threats.

- [ ] Defenders rely solely on signature-based detection.
- [x] Machine learning models are employed by both attackers and defenders.

Nice work

Correct! Both attackers and defenders utilize machine learning models to enhance their strategies and adapt to new threats.

- [ ] Attackers reuse old malware without modification.

5. What characteristic most clearly defines polymorphic malware, making it challenging for traditional detection methods? **1 / 1 point**

- Malware that replicates itself without modifying its code.
- Malware that employs encryption to hide its payload.
- Malware that simultaneously targets multiple operating systems.
- Malware that alters its code to evade detection

Nice work

Correct! Polymorphic malware modifies its code to evade detection by signature-based methods, making identification difficult.